

Arvin Zaker

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Education

Doctor of Medicine (MD) <i>University of British Columbia (UBC), Vancouver, BC</i>	Aug 2025–Present
Honours Bachelor of Science, Translational and Molecular Medicine <i>University of Ottawa, Ottawa, ON; cumulative grade point average: 9.98/10</i>	Sep 2019–Apr 2023

Research Experience

Medical Imaging Artificial Intelligence <i>University of British Columbia, Vancouver, BC</i> <i>Supervisor: Dr. Ilker Hacihaliloglu</i> - Curated 14 datasets and built a Python pipeline to train 1,176 models on 38,357 paired mammograms. - Externally tested selected models on EMBED dataset to identify AI model performance failures and its association with breast density, race, ethnicity, and histology.	Feb 2026–Present
E-Cigarette Flavour Policy Evaluation <i>University of British Columbia, Vancouver, BC</i> <i>Supervisor: Dr. Kevin Duan</i> Compared the policy effects of comprehensive e-cigarette flavour bans and specialty-retailer restrictions among 1,516 youth across 5 Canadian provinces using CTNS data and survey-weighted difference-in-differences.	Feb 2026–Present
EpiGenomics and Chromatin Biology 🧬 <i>University of Ottawa, Ottawa, ON</i> <i>Supervisor: Dr. Arvind Mer</i> - Analyzed clinical exosomal DNA to identify differentially methylated regions and pathways and integrated multi-omics evidence from 44 studies and databases. - Built a reproducible CUT&RUN workflow for processing, quality control, annotation, and collaborator use. - Aggregated and visualized CXXC1 mutational data across the genome and plant homeodomain (PHD).	Mar 2024–Jan 2026
Text-Mining Feature Selection for Anticancer Drug Response 🧬 <i>University of Ottawa, Ottawa, ON</i> <i>Supervisor: Dr. Arvind Mer</i> - Devised and benchmarked text-mining feature-selection methods across drug-response prediction models and validation datasets. - Built reproducible local and high-performance computing workflows and visualizations for model evaluation.	Sep 2023–Feb 2024
Bacterial Genomics <i>University of Ottawa, Ottawa, ON</i> <i>Supervisors: Dr. Arvind Mer and Dr. Vanessa M. D'Costa</i> Built genomic annotation and phylogenetic-analysis workflows and characterized type IV secretion system clusters across 118 <i>Acinetobacter</i> bacterial and plasmid strains.	Jul 2023–Feb 2024
SMC5/6 Cancer Phenogenomics 🧬 <i>University of Ottawa, Ottawa, ON</i> <i>Supervisors: Dr. Arvind Mer and Dr. Damien D'Amours</i> Analyzed SMC5/6 alterations, genomic instability, transcriptomic changes, cancer phenotypes, and survival in a pan-cancer cohort of more than 65,000 patients across 144 studies.	May 2022–Aug 2022
Optogenetic Neurophysiology <i>University of Ottawa, Ottawa, ON</i> <i>Supervisors: Dr. Arvind Mer and Dr. Greg Silasi</i> Co-developed software to quantify hemodynamic and cortical responses to optogenetic stimulation after stroke <i>in vivo</i>.	Mar 2022–Sep 2024
Patient-Derived Xenograft Drug-Response Modelling <i>University of Ottawa, Ottawa, ON</i> <i>Supervisor: Dr. Arvind Mer</i> - Developed the open-source xeMetron package to calculate 250 measures of patient-derived xenograft chemotherapy response. - Built machine-learning models to identify biomarkers of drug response and quantify anti-cancer drug response in clinical-trial data.	Jan 2022–June 2026

Bacteriophage Discovery

University of Ottawa, Ottawa, ON

Supervisor: Dr. Adam Rudner

- Isolated, amplified, characterized, and annotated 2 novel bacteriophages, Arzan and Khorshid.

Sep 2021–Apr 2022

Muscle Physiology

University of Ottawa, Ottawa, ON

Supervisor: Dr. Jean-Marc Renaud

- Designed electrophysiology experiments on adenosine receptors and skeletal-muscle fatigue and performed tissue staining, fluorescence imaging, and quantitative image analysis.

Nov 2020–Jul 2022

Peer-Reviewed Publications

- Asare-Werehene M, **Zaker A**, Tripathi S, Communal L, Carmona E, Mes-Masson A-M, Tsang BK, Mer A. Spotlight on nuclear PD-L1 in ovarian cancer chemoresistance: hidden but mighty. *Front Immunol.* 2025;16. doi:10.3389/fimmu.2025.1543529.
- Grégoire S, Grégoire J, Yang Y, Capitani S, Joshi M, Sarvan S, **Zaker A**, Ning Z, Figeys D, Ulrich K, Brunzelle JS, Mer A, Couture J-F. Structural insights into an atypical histone binding mechanism by a PHD finger. *Structure.* 2024;32(9):1498–1506.e4. doi:10.1016/j.str.2024.06.017.
- Wu G, **Zaker A** (co-first author), Ebrahimi A, Tripathi S, Mer AS. Text-mining-based feature selection for anticancer drug response prediction. *Bioinform Adv.* 2024;4(1):vbae047. doi:10.1093/bioadv/vbae047.
- Nasser F, Gaudreau A, Lubega S, **Zaker A**, Xia X, Mer AS, D'Costa VM. Characterization of the diversity of type IV secretion system-encoding plasmids in *Acinetobacter*. *Emerg Microbes Infect.* 2024;13(1):2320929. doi:10.1080/22221751.2024.2320929.
- Roy S, **Zaker A**, Mer A, D'Amours D. Large-scale phenogenomic analysis of human cancers uncovers frequent alterations affecting SMC5/6 complex components in breast cancer. *NAR Cancer.* 2023;5(3):zcad047. doi:10.1093/narcan/zcad047.

Presentations, Posters, Seminars & Abstracts

- Poster and oral presentations; presenting author.** Roy S, **Zaker A**, Mer A, D'Amours D. Large-scale phenogenomic analysis of human cancers uncovers frequent alterations affecting SMC5/6 complex components in breast cancer. *Canadian Society for Molecular Biosciences International Conference*; May 2023; Ottawa, ON.
- Oral presentation; presenting author.** **Zaker A**, Mer A. Comparative analysis of drug response metrics in patient-derived xenografts. *4th Annual University of Ottawa Faculty of Medicine Research Day*; Sep 2022; Ottawa, ON. Panel question period; 18 students were selected.
- Abstract and presentation; co-presenter.** Featherstone A, Radar A, Salama A, Tiukuvaara S, Ferri O, **Zaker A**, et al. Give us a millet of your time: dehusking the singleton bacteriophage Arzan. *SEA Symposium*; Apr 2022; Pittsburgh, PA.
- Abstract and presentation; co-presenter.** Friesen R, Algharbi S, **Zaker A**, Mahdi O, Gao R, Ferri O, et al. Investigating nucleotide-binding proteins in bacteriophage JohnDoe. *SEA Symposium*; Apr 2022; Pittsburgh, PA.
- Abstract and presentation; co-presenter.** Cildir Z, Abdisalam S, Alam A, **Zaker A**, Toex N, Chandra A, et al. Prophage liberation! A method to increase the titer of the temperate phages Khorsid and Winseler. *SEA Symposium*; Apr 2022; Pittsburgh, PA.
- Seminar; presenter.** **Zaker A**, Mer A. Comparative analysis of drug response metrics in patient-derived xenografts. *University of Ottawa Translational and Molecular Medicine Seminar*; Apr 2022. Twenty-minute presentation and question period.
- Poster presentation; presenting author.** **Zaker A**, Mer A. Analysis of drug response in patient-derived xenografts. *Faculty of Medicine, University of Ottawa*; Feb 2022; Ottawa, ON.
- Poster and oral presentations; presenting author.** **Zaker A**, McCarthy L, Jung D, Karunakaran G, Rudner A. Discovery and characterization of bacteriophage Arzan. *University of Ottawa Faculty of Medicine*; Dec 2021; Ottawa, ON.
- Abstract and poster presentation; presenting author.** **Zaker A**, McRae C, Renaud JM. Role of adenosine receptors in mouse skeletal-muscle fatigue kinetics. *University of Ottawa Undergraduate Research Opportunity Program Symposium*; Mar 2021.

Leadership & Governance

Research and Innovation Lead Jul 2026–Present

MD Student Learning Group, University of British Columbia, Vancouver, BC

- Lead the research and innovation portfolio, connecting peers with learning opportunities and promoting evidence-informed practice, emerging technologies, and student-led scholarship.

Vice-President, External Relations Jul 2026–Present

Artificial Intelligence in Medicine Society, University of British Columbia, Vancouver, BC

- Helped launch a student community focused on responsible artificial intelligence in health care and build external partnerships for panels, workshops, and collaborative projects.

Chief Technology Officer Jun 2026–Present

Learn Freely, nonprofit organization, Vancouver, BC

- Lead development of a learning management system and workflows for tutoring assignments, progress tracking, project management, and organizational collaboration.

Board Member Jun 2026–Present

Inclusion, Diversity and Equity Advisory Committee, Doctors of BC, Vancouver, BC; three-year term

- Selected to advise the Board of Directors on equity, diversity, and inclusion across governance bodies, members, and the medical profession.

Founder and President Jun 2020–Sep 2024

Hands on Health Magazine, University of Ottawa, Ottawa, ON

- Founded and led a 20+ member team that published 50+ faculty-reviewed, evidence-based health articles.
- Built editorial, review, and communication workflows that sustained publication during remote operations.

Founder and President Dec 2020–Apr 2023

Canadians Sharing Locally and Globally uOttawa, University of Ottawa, Ottawa, ON

- Partnered with a local charity to help raise more than \$100,000 for personal protective equipment and ventilators for rural hospitals in India.
- Led advocacy events and social media outreach on global health inequities and social determinants of health.

Teaching & Mentorship

Founder and Production Designer Apr 2026–Present

UBC Integrated Anatomy Atlas, University of British Columbia, Vancouver, BC

- Collaborated with UBC anatomy faculty to develop a comprehensive flashcard deck spanning both preclinical years of the UBC MD curriculum.
- Produced anatomy images and 320 structure-identification questions across 14 laboratory modules in Anki, study-site, and simulated bell-ringer formats.
- Integrated the resource into UBC Elentra, making it available to all UBC medical students.

Premedical Student Mentor Aug 2025–Present

Supporting Applications, Supporting Interviews (SASI), Community of Support, Temerty Faculty of Medicine, University of Toronto, Toronto, ON

- Mentored 3 premedical students from groups underrepresented in medicine through one-on-one meetings, individualized application strategy and feedback on interview and written submissions.

Science, Technology, Engineering, and Mathematics Tutor Sep 2023–Jul 2025

Oxford Learning Oak Ridges and Inspiration Learning, Toronto, ON

- Tutored 20+ students from grades 1–12 and university across biology, chemistry, physics, and mathematics using individualized learning plans.

Peer Tutor and Mentor Sep 2018–Present

- Guided 50+ middle-school and high-school students in science and mathematics and supported transitions to university through academic guidance and peer mentorship.

Community Engagement

First-Year Representative Sep 2025–Sep 2026

Persian Medical Society, University of British Columbia, Vancouver, BC

- Help create and promote peer-support, mentorship, and cultural events for Persian-identifying students and allies, strengthening community connection and inclusive representation within the medical school.

Administrative Receptionist **Sep 2022–Jun 2023**
Chadha Medical Clinic, Ottawa, ON

- Coordinated appointments, referrals, and screenings in a primary-care practice serving approximately 5,000 patients and managed 60+ patient contacts weekly.

Volunteer Coordinator and Co-Chair **Sep 2020–Sep 2024**
Best Buddies uOttawa, University of Ottawa, Ottawa, ON

- Recruited and paired 50+ student volunteers with people with intellectual and developmental disabilities.
- Led accessible events and revised recruitment to address audiovisual and language barriers identified through feedback.

Honours, Scholarships & Awards

Florence E. Heighway Summer Research Award **Jun 2026–Aug 2026**
Faculty of Medicine Summer Student Research Program, University of British Columbia, Vancouver, BC

- Awarded 8 weeks of funding for a mammography artificial-intelligence failure-analysis project.

Third Place, IFHP NavAID Challenge **Jun 2026**
University of British Columbia, Vancouver, BC

- Collaborated with 2 medical students to develop a 75-minute refugee-health case with 4 decision points and 7 interprofessional objectives.

Undergraduate Student Research Award (USRA) **May 2023–Aug 2023**
Natural Sciences and Engineering Research Council of Canada (NSERC), University of Ottawa, Ottawa, ON

- Awarded \$7,500 for a 3-month research project based on academic merit and research-proposal quality; one of 88 awards at the University of Ottawa.

Translational and Molecular Medicine Summer Training Award **May 2022–Aug 2022**
University of Ottawa, Ottawa, ON

- Awarded \$8,900 for a 3-month supervised research project based on academic merit and research-proposal quality.

Work in Biomedical Research Scholarship **May 2022–Aug 2022**
University of Ottawa, Ottawa, ON

- Awarded \$3,500 for a 3-month research project through a competition with a success rate below 50%.

Undergraduate Research Opportunity Program Award **Dec 2020**
University of Ottawa, Ottawa, ON

- Awarded \$1,000 for a 4-month supervised research project based on academic merit and research-proposal quality.

Dean's Honour List and Merit Scholarship, **Sep 2019–Apr 2023**
University of Ottawa, Ottawa, ON

Projects & Interests

Explainable Heart-Disease Risk Modelling  **Mar 2026**
National Canadian Medical Datathon, Vancouver, BC

- Collaborated with 5 colleagues to integrate 920 records from 4 University of California, Irvine heart-disease cohorts and develop a regularized logistic-regression model.
- Built CLEAR, an R Shiny demonstration tool providing explainable and individualized risk estimates using SHapley Additive exPlanations and plain-language feature summaries.

Clinical Bioinformatics Reproducibility Project  **Mar 2023**

- Reproduced the bioinformatics analysis reported by Viñas et al. (2020; PMID: 32662516) and produced a documented report, presentation, and explanatory visualizations.

Systems Administration and Research Computing **Jan 2022–Present**

- Built and maintain a 16-terabyte Unraid server for centralized family storage, automated backups, data recovery, and Docker-based web services.
- Built and administer 3 Linux workstations for machine learning, and high-performance computing.